

HDOC – Day 21

C Programming Assignment

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Menu-Driven Operations on Integers

Algorithm • Test Cases • C Program • Compilation & Execution Outputs

1. Problem Statement

Write a menu-driven C program that accepts two positive integers m and n , where $m \leq n$. The program repeatedly displays a menu and uses a switch statement to perform multiplication-table, addition-table, subtraction-table, and division-table operations for every integer from m to n . Division results are displayed to two decimal places. The program continues until the user chooses Exit and reports an error for an invalid menu choice.

2. Algorithm

1. Start.
2. Read two integers m and n .
3. If $m \leq 0$, $n \leq 0$, or $m > n$, display an invalid-input message and stop.
4. Display the menu with choices 1 to 5 and read choice.
5. Use switch(choice):
 6. Choice 1: For each number from m to n , print number $\times i$ for $i = 1$ to 10.
 7. Choice 2: For each number from m to n , print number $+ i$ for $i = 1$ to 10.
 8. Choice 3: For each number from m to n , print number $- i$ for $i = 1$ to 10.
 9. Choice 4: For each number from m to n , print number $/ i$ for $i = 1$ to 10 using floating-point division and `%.2f`.
10. Choice 5: Display the exit message.
11. Default: Display an invalid-menu-choice message.
12. Repeat the menu until choice = 5.
13. Stop.

3. Test-Case Table

No.	Input/Choice	Expected Result	Status
1	m=2, n=3; choice=1	Multiplication tables of 2 and 3 (1 to 10), then Exit	Pass
2	m=2, n=2; choice=2	Addition table of 2 using addends 1 to 10, then Exit	Pass
3	m=5, n=5; choice=3	Subtraction table of 5 using subtrahends 1 to 10, then Exit	Pass
4	m=7, n=7; choice=4	7/1 through 7/10, rounded to two decimals, then Exit	Pass
5	m=2, n=2; choice=9	Invalid menu choice message; menu appears again; choice 5 exits	Pass

4. C Program

```
#include <stdio.h> int main(void) { int m, n, choice, num, i;
printf("Enter two positive integers m and n (m <= n): "); if
(scanf("%d %d", &m, &n) != 2 || m <= 0 || n <= 0 || m > n) {
printf("Invalid input. Enter positive integers with m <= n.\n");
return 0;
}
do {
printf("\nMENU\n");
printf("1. Print first 10 multiples\n");
printf("2. Print addition tables\n");
printf("3. Print subtraction tables\n");
printf("4. Print division results\n");
printf("5. Exit\n"); printf("Enter your
choice: "); scanf("%d", &choice);
switch (choice) { case 1: for (num = m;
num <= n; num++) {
printf("\nMultiples of %d:\n", num);
for (i = 1; i <= 10; i++)
printf("%d x %d = %d\n", num, i, num * i);
}
```

```
break; case 2: for (num = m; num <= n;
num++) { printf("\nAddition table of
%d:\n", num); for (i = 1; i <= 10; i++)
printf("%d + %d = %d\n", num, i, num + i);
}
break; case 3: for (num = m; num <= n;
num++) { printf("\nSubtraction table of
%d:\n", num); for (i = 1; i <= 10; i++)
printf("%d - %d = %d\n", num, i, num - i);
}
break; case 4: for (num = m; num <= n;
num++) { printf("\nDivision results for
%d:\n", num);
for (i = 1; i <= 10; i++)
printf("%d / %d = %.2f\n", num, i, (double)num / i);
}
break; case 5: printf("Exiting
program.\n"); break; default:
printf("Invalid menu choice. Please enter 1 to 5.\n");
}
} while (choice != 5); return
0; }
```

OUTPUTS

Test Case 1-Multiplication Table

```
Session Actions Edit View Help
(kali@kali)-[~]
$ ./a.out
Enter two positive integers m and n (m ≤ n): 2 3

MENU
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division results
5. Exit
Enter your choice: 1

Multiples of 2:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Multiples of 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

MENU
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division results
5. Exit
Enter your choice: 5
Exiting program.
```

Test Case 2 -Addition Table

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter two positive integers m and n ( $m \leq n$ ): 2 2  
  
MENU  
1. Print first 10 multiples  
2. Print addition tables  
3. Print subtraction tables  
4. Print division results  
5. Exit  
Enter your choice: 2  
  
Addition table of 2:  
2 + 1 = 3  
2 + 2 = 4  
2 + 3 = 5  
2 + 4 = 6  
2 + 5 = 7  
2 + 6 = 8  
2 + 7 = 9  
2 + 8 = 10  
2 + 9 = 11  
2 + 10 = 12  
  
MENU  
1. Print first 10 multiples  
2. Print addition tables  
3. Print subtraction tables  
4. Print division results  
5. Exit  
Enter your choice: 5  
Exiting program.
```

Test Case 3- Subtraction Table

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter two positive integers m and n ( $m \leq n$ ): 5 5  
  
MENU  
1. Print first 10 multiples  
2. Print addition tables  
3. Print subtraction tables  
4. Print division results  
5. Exit  
Enter your choice: 3  
  
Subtraction table of 5:  
5 - 1 = 4  
5 - 2 = 3  
5 - 3 = 2  
5 - 4 = 1  
5 - 5 = 0  
5 - 6 = -1  
5 - 7 = -2  
5 - 8 = -3  
5 - 9 = -4  
5 - 10 = -5  
  
MENU  
1. Print first 10 multiples  
2. Print addition tables  
3. Print subtraction tables  
4. Print division results  
5. Exit  
Enter your choice: 5  
Exiting program.
```

Test Case 4 -Division Table

```
(kali@kali)-[~]
$ ./a.out
Enter two positive integers m and n (m ≤ n): 7 7

MENU
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division results
5. Exit
Enter your choice: 4

Division results for 7:
7 / 1 = 7.00
7 / 2 = 3.50
7 / 3 = 2.33
7 / 4 = 1.75
7 / 5 = 1.40
7 / 6 = 1.17
7 / 7 = 1.00
7 / 8 = 0.88
7 / 9 = 0.78
7 / 10 = 0.70

MENU
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division results
5. Exit
```

Test Case 5-Invalid choice

```
(kali@kali)-[~]
$ ./a.out
Enter two positive integers m and n (m ≤ n): 2 2

MENU
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division results
5. Exit
Enter your choice: 9
Invalid menu choice. Please enter 1 to 5.

MENU
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division results
5. Exit
Enter your choice: 5
Exiting program.
```

6. Conclusion

The menu-driven C program was implemented using a switch statement and loops. It correctly prints multiplication, addition, subtraction, and division tables for all integers from m to n, validates invalid menu choices, formats division results to two decimal places, and repeats until the Exit option is selected. All prepared test cases passed.